

RAG Practice Manual

This synthetic multi-page PDF is designed for extraction, page metadata, chunking, and indexing exercises.

Page section 1: This paragraph intentionally repeats the topic with additional context for extraction testing. This synthetic multi-page PDF is designed for extraction, page metadata, chunking, and indexing exercises.

Retrieval

Retrieval selects potentially useful passages from a document collection. Semantic retrieval uses embeddings while lexical retrieval uses terms and ranking functions.

Page section 2: This paragraph intentionally repeats the topic with additional context for extraction testing. Retrieval selects potentially useful passages from a document collection. Semantic retrieval uses embeddings while lexical retrieval uses terms and ranking functions.

Chunking

Chunking divides documents into smaller units. Different strategies have different trade-offs in context preservation, chunk size, and retrieval precision.

Page section 3: This paragraph intentionally repeats the topic with additional context for extraction testing. Chunking divides documents into smaller units. Different strategies have different trade-offs in context preservation, chunk size, and retrieval precision.

Embeddings

Embeddings map text to vectors. Similar meanings can be compared using cosine similarity or another distance metric.

Page section 4: This paragraph intentionally repeats the topic with additional context for extraction testing. Embeddings map text to vectors. Similar meanings can be compared using cosine similarity or another distance metric.

Vector Databases

Vector databases persist vectors and associated metadata and expose similarity-search operations.

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RAG Generation

The retrieved passages are added to the model context. The model can then answer using the supplied evidence.

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Evaluation

Evaluation datasets can contain questions, expected answers, keywords, and source identifiers. They support repeatable experiments.

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Deployment

A production service can use FastAPI for its backend and Streamlit for its user interface. Configuration and secrets should be managed separately from code.

Page section 8: This paragraph intentionally repeats the topic with additional context for extraction testing. A production service can use FastAPI for its backend and Streamlit for its user interface. Configuration and secrets should be managed separately from code.